

☒ Easy Level (10 Questions)

1.

Which command displays current CPU and memory usage in real-time?

- A) df
- B) top
- C) du
- D) ifconfig

Ans: B

☒ top shows live CPU, memory, and process info.

2.

Which file is used to make kernel parameter changes persistent?

- A) /etc/init.d/rc.local
- B) /etc/sysctl.conf
- C) /etc/modules.conf
- D) /etc/hosts

Ans: B

☒ sysctl.conf stores persistent sysctl settings.

3.

What does the command `df -h` show?

- A) File permissions
- B) Running processes
- C) Disk space usage
- D) Memory stats

Ans: C

☒ df -h shows disk usage in human-readable format.

4.

Which protocol is most commonly exploited in a DDoS attack?

- A) HTTP
- B) SMTP
- C) FTP
- D) SSH

Ans: A

☒ DDoS often targets HTTP (port 80) to flood websites.

5.

What does 'RAM' stand for?

- A) Read Access Memory
- B) Random Access Memory
- C) Remote Access Memory
- D) Random Available Memory

Ans: B

6.

What is the goal of performance tuning?

- A) Install updates
- B) Improve system speed and efficiency
- C) Fix bugs
- D) Add new hardware

Ans: B

7.

Which tool helps track network usage and open connections?

- A) netstat
- B) fdisk
- C) ps
- D) whoami

Ans: A

8.

Which log file contains kernel messages?

- A) /var/log/auth.log

- B) /var/log/kern.log
- C) /var/log/syslog
- D) /etc/kernel.log

Ans: B

9.

What is the purpose of threat modeling?

- A) Encrypt data
- B) Predict and prevent attacks
- C) Set permissions
- D) Format disk

Ans: B

10.

Which is a valid protection method?

- A) Root login over telnet
- B) Weak passwords
- C) Role-Based Access Control
- D) Disabled firewall

Ans: C

☒ Medium Level (20 Questions)

11.

Which command shows memory usage statistics?

- A) iostat
- B) vmstat
- C) ping
- D) netstat

Ans: B

☒ vmstat displays memory, swap, and CPU info.

12.

Which file should be checked for service failure logs in a systemd system?

- A) /etc/sysctl.conf
- B) /var/log/dmesg
- C) journalctl
- D) /etc/logrotate.conf

Ans: C

13.

Which component of STRIDE refers to unauthorized elevation of privileges?

- A) Spoofing
- B) Elevation of Privilege
- C) Repudiation
- D) Denial of Service

Ans: B

14.

Which of the following is a kernel parameter that controls swap behavior?

- A) fs.swappiness
- B) mem.swappiness
- C) vm.swappiness
- D) proc.swappiness

Ans: C

15.

How would you reload sysctl settings without rebooting?

- A) sysctl --reload
- B) sysctl -a
- C) sysctl -p
- D) reload sysctl

Ans: C

16.

Which command helps find processes using the most CPU?

- A) find
- B) du
- C) top
- D) stat

Ans: C

17.

Which is NOT a phase of threat modeling?

- A) Identify assets
- B) Locate vulnerabilities
- C) Format the disk
- D) Develop mitigations

Ans: C

18.

What does the ping command test?

- A) Disk speed
- B) DNS lookup
- C) Network connectivity
- D) User permissions

Ans: C

19.

Which command shows file system usage by directory?

- A) df
- B) du
- C) ls
- D) touch

Ans: B

20.

What is the role of DREAD in security?

- A) Detect malware
- B) Categorize threats
- C) Secure database
- D) Log user actions

Ans: B

21.

Which of the following is NOT part of the DREAD model?

- A) Damage
- B) Reproducibility
- C) Detectability
- D) Duration

Ans: D

22.

Which command shows I/O statistics?

- A) iostat
- B) free
- C) who
- D) cal

Ans: A

23.

What's the main cause of high load average in Linux?

- A) Too many idle processes
- B) Insufficient swap
- C) High CPU usage
- D) Large number of open ports

Ans: C

24.

Which tool is used to analyze kernel events and bottlenecks?

- A) sysctl
- B) perf
- C) cal
- D) uname

Ans: B

25.

What is an effective method to detect anomalies in system logs?

- A) head
- B) tail
- C) Log monitoring with regular expressions
- D) Deleting logs

Ans: C

26.

Which is a valid method to maintain system performance over time?

- A) Never update
- B) Use weak passwords

C) Log rotation and cleanup

D) Run as root always

Ans: C

27.

Which of these is a physical security method?

A) Role-based access

B) TLS encryption

C) Locked server room

D) Antivirus

Ans: C

28.

In threat modeling, 'attack vector' refers to:

A) Anticipated defense strategy

B) Firewall configuration

C) The path or method used to exploit a vulnerability

D) DNS settings

Ans: C

29.

Which setting is often tuned to improve disk I/O?

- A) vm.swapiness
- B) block size
- C) host name
- D) syslog level

Ans: B

30.

What is journalctl -xe used for?

- A) Export logs
- B) View error and critical logs
- C) Email logs
- D) Reset journal

Ans: B

☒ Hard Level (10 Questions)

31.

Which combination is most effective for identifying root causes in Linux crashes?

- A) ping + traceroute

- B) dmesg + journalctl + logs
 - C) ps + who
 - D) top + cal
- Ans: B**
-

32.

What does perf top do?

- A) Shows network traffic
 - B) Displays top disk usage
 - C) Shows live kernel-level profiling
 - D) Stops running services
- Ans: C**
-

33.

In which file do you define custom sysctl values in CentOS/RHEL?

- A) /boot/grub.conf
 - B) /etc/sysctl.d/custom.conf
 - C) /etc/modprobe.conf
 - D) /etc/profile
- Ans: B**
-

34.

Which STRIDE element deals with fake identity usage?

- A) Spoofing
- B) Repudiation
- C) Tampering
- D) DoS

Ans: A

35.

Which command is used to find a memory leak in real-time?

- A) free
- B) valgrind
- C) netstat
- D) whoami

Ans: B

36.

What kernel parameter controls the TCP connection backlog?

- A) net.core.tcp_syncookies
- B) net.core.netdev_max_backlog
- C) net.ipv4.tcp_max_syn_backlog
- D) net.ipv4.tcp_sync_backlog

Ans: C

37.

Which metric best reflects system responsiveness?

- A) CPU usage
- B) Disk I/O
- C) Load average over 1 minute
- D) Number of open files

Ans: C

38.

What is an IDS?

- A) Internet Directory Service
- B) Intrusion Detection System
- C) Internal Disk Scanner
- D) Integrated Data Storage

Ans: B

39.

Why would you use auditd?

- A) Monitor security violations and user actions
- B) Check disk usage
- C) Upgrade Linux kernel
- D) Start Apache

Ans: A

40.

Which of the following can help reduce DDoS impact?

- A) Disabling SSH
- B) Setting vm.swappiness to 100
- C) Using a reverse proxy or WAF
- D) Blocking ICMP

Ans: C